

Digital-Twin.md



Digital Twin

ArenaMind AI UI/UX Specification

Screen ID: DT-001

Version: 1.0

Status: Draft

Priority: ★★★★★

Overview

The **Digital Twin** is the core intelligence layer of ArenaMind AI.

Instead of displaying operational data through traditional dashboards, ArenaMind AI visualizes the stadium as a **living digital ecosystem**.

The Digital Twin continuously synchronizes operational information from multiple sources into a unified spatial interface.

It enables users to:

- Monitor live stadium conditions
- Predict operational risks
- Simulate future scenarios
- Coordinate incident response
- Understand crowd behavior
- Optimize stadium operations

The Digital Twin is **the primary operational workspace** rather than an optional visualization.

Product Vision

| **"One interactive stadium. Every operational insight."**

ArenaMind AI transforms a static stadium into an intelligent digital replica capable of reasoning about the physical environment.

The Digital Twin acts as the operational mirror of the real venue.

Primary Users

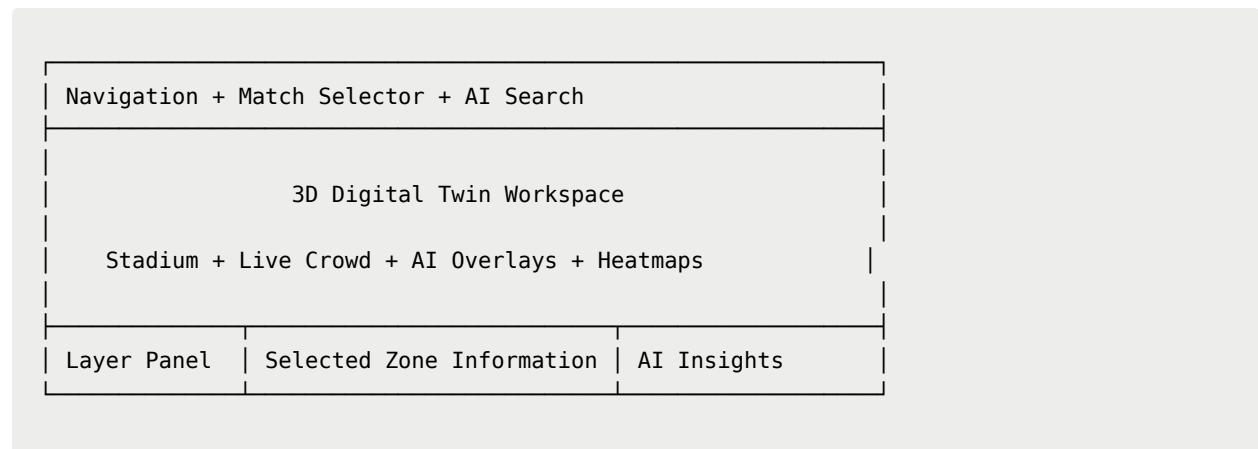
- Stadium Operations Manager
 - Security Teams
 - Emergency Response
 - Volunteer Coordinator
 - Medical Operations
 - Transport Authorities
 - Sustainability Team
 - FIFA Match Officials
-

User Goals

Users should be able to answer:

- Where is the issue?
 - Why is it happening?
 - What will happen next?
 - Which resources are available?
 - What should we do immediately?
-

Screen Layout



Core Objectives

The Digital Twin should provide

- ✓ Real-time visualization
- ✓ Predictive intelligence
- ✓ Interactive exploration
- ✓ Operational simulations
- ✓ AI-assisted decision support

Stadium Model

The center of the interface is a highly detailed 3D stadium model.

Supported interactions

- Rotate
- Zoom
- Pan
- Focus
- Reset Camera

- Fullscreen

Users should feel as though they are inspecting the real stadium.

Interactive Zones

Every operational area is interactive.

Stadium

- |— Gates
- |— Seating
- |— VIP
- |— Hospitality
- |— Food Courts
- |— Parking
- |— Medical
- |— Security
- |— Volunteer Base
- |— Broadcast
- |— Media
- |— Emergency Exit
- |— Power Systems
- |— Water Systems
- |— Control Rooms

Every zone supports

Hover

↓

Click

↓

Inspect

↓

AI Analysis

Zone Information Panel

Selecting a zone opens an operational overview.

Example

Gate B

Status

Operational

Current Occupancy

81%

Capacity

95%

Prediction

High congestion in 12 min

Confidence

96%

Recommendations

Open Gate D

Deploy 4 Volunteers

Broadcast multilingual guidance

Layer Control

Users can toggle operational layers.

☒ Crowd

☒ Security

☒ Medical

☒ Volunteers

- ☑ Weather
- ☑ Transport
- ☑ Parking
- ☑ Energy
- ☑ Accessibility
- ☑ Utilities
- ☑ CCTV
- ☑ Emergency
- ☑ Sustainability

Multiple layers may be active simultaneously.

Crowd Intelligence Layer

Visualized using animated particles.

Density

Green

↓

Yellow

↓

Orange

↓

Red

Each particle represents aggregated crowd movement.

AI projects future crowd flow using translucent animated paths.

Transportation Layer

Displays

- Shuttle buses
- Metro stations
- Parking occupancy
- Ride-sharing zones
- Walking routes
- Traffic conditions

Vehicles move continuously along predefined routes.

Volunteer Layer

Displays live volunteer positions.

Each volunteer marker includes

- Name
- Role
- Assigned Task
- Current Location
- ETA
- Availability

Users can assign new tasks directly from the map.

Security Layer

Displays

- CCTV coverage
- Restricted zones
- Security checkpoints

- Patrol routes
- Active incidents

High-risk zones pulse in red.

Medical Layer

Visualizes

- Medical stations
- Ambulances
- First-aid volunteers
- AED locations
- Patient incidents

Selecting a medical event opens:

Patient Status

Nearest Medical Team

ETA

Recommended Route

Hospital Availability

Accessibility Layer

Displays

- Wheelchair routes
- Accessible entrances
- Elevators
- Hearing assistance

- Quiet zones
- Accessible seating

Routes adapt dynamically during incidents.

Sustainability Layer

Visualizes

- Energy usage
- Water consumption
- HVAC activity
- Waste collection
- Carbon emissions

AI highlights optimization opportunities.

Weather Layer

Displays

- Rain
- Wind
- Temperature
- Lightning
- Visibility

Weather events affect

- Crowd movement
 - Route recommendations
 - Transportation
 - Emergency planning
-

Emergency Layer

Critical incidents trigger spatial animations.

Example

Fire Alarm

↓

Affected zone flashes red

↓

Emergency exits illuminate

↓

Crowd rerouting displayed

↓

Medical routes appear

↓

Volunteer deployment visualized

↓

AI recommendations generated

AI Insight Panel

Persistent right-side panel.

Displays

Operational Summary

Current Risk Level

Predicted Events

Confidence Score

Recommended Actions

Estimated Impact

AI continuously updates this panel.

AI Scenario Simulator

Users can simulate future events.

Examples

Close Gate A

Heavy Rain

Medical Emergency

Power Failure

Transport Delay

Security Breach

ArenaMind AI simulates

- Crowd redistribution
- Queue changes
- Travel delays
- Volunteer reassignment
- Energy impact
- Emergency response

Results appear instantly.

Timeline Controls

Users can scrub through time.

Past

↓

Present

↓

+5 min

↓

+10 min

↓

+30 min

↓

+1 hr

Predictions animate directly on the Digital Twin.

Camera System

Modes

Overview

Inspection

Emergency

AI Focus

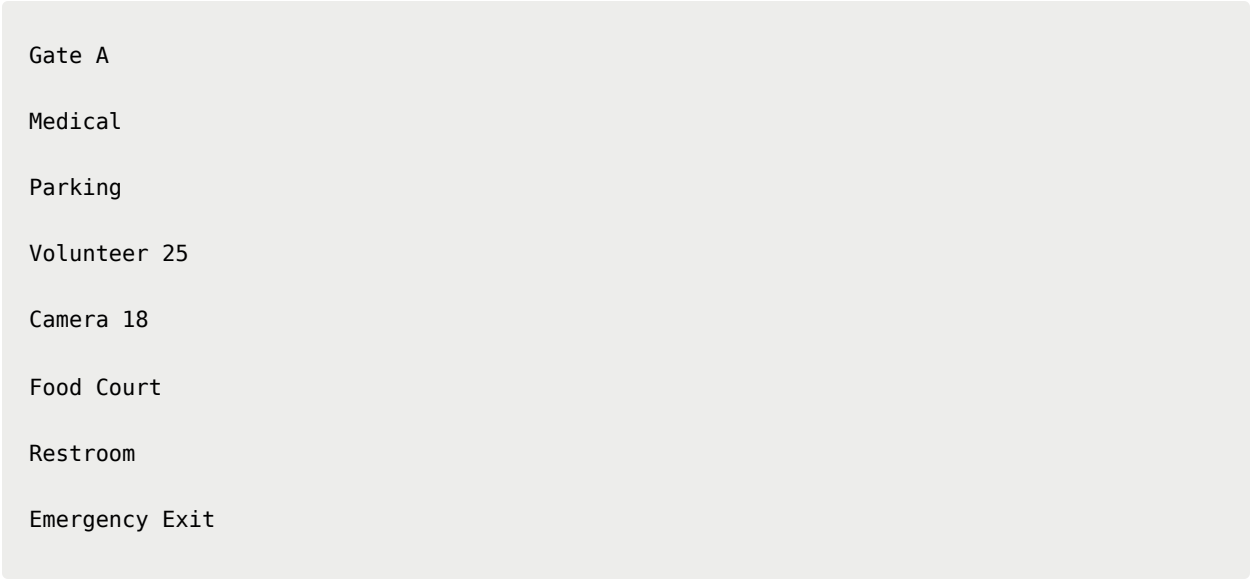
Presentation

Free Camera

Smooth transitions only.

Search

Users may search



Gate A
Medical
Parking
Volunteer 25
Camera 18
Food Court
Restroom
Emergency Exit

Camera automatically navigates.

Context Menu

Right-clicking a zone opens

- Inspect
 - View Cameras
 - Generate Report
 - Assign Volunteers
 - Broadcast Message
 - Open Incident
 - Simulate Scenario
-

AI Interaction

Example

User

Why is Gate C crowded?

ArenaMind AI

Current occupancy is 86%.

Reason:

- Shuttle arrival
- Food court demand
- Rain near Gate A

Prediction:

Queue will increase by 18%.

Recommendation:

Redirect spectators to Gate D.

Notification Integration

Digital Twin responds instantly.

Examples

- Camera anomaly
- Medical alert
- Volunteer request
- Weather warning
- Transport delay

Affected zones animate immediately.

Motion Design

Hover



Glow



Elevation

Selection



Camera zoom



Panel expansion

Incident



Pulse



Heatmap



Lighting change

Accessibility

Supports

- Keyboard navigation
 - Voice interaction
 - Screen readers
 - High contrast
 - Reduced motion
 - Zoom controls
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Responsive Design

Desktop

Full 3D experience.

Tablet

Simplified Digital Twin.

Reduced particle count.

Mobile

Interactive map mode.

3D preview.

Zone inspection.

AI assistant.

Empty States

No active incidents.

Digital Twin synchronized.

All systems operating normally.

Loading Experience

Instead of

Loading...

Display

- Stadium wireframe

- Progressive model loading
 - AI synchronization animation
 - Layer initialization
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Performance Targets

Metric	Target
Model Load	<3 sec
Interaction Latency	<100 ms
Frame Rate	60 FPS
Preferred Frame Rate	120 FPS
Layer Toggle	<200 ms

Technical Stack

Rendering

- Three.js
- React Three Fiber
- Drei

Animation

- GSAP
- Framer Motion

Data

- Firebase
- Firestore
- WebSockets

Maps

- Google Maps Platform

AI

- Gemini 2.5 (Primary)
- Groq (Fallback)
- Vertex AI
- RAG Engine

Optimization

- LOD (Level of Detail)
 - Geometry Instancing
 - Lazy Loading
 - Suspense
 - Draco Compression
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Acceptance Criteria

- Stadium model loads interactively.
 - All operational layers can be toggled independently.
 - AI insights update automatically.
 - Zone selection displays contextual information.
 - AI Scenario Simulator produces visual predictions.
 - Gemini fallback to Groq occurs seamlessly.
 - Camera transitions remain smooth.
 - Performance meets 60 FPS target.
 - Accessible via keyboard and screen readers.
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Future Enhancements

- Multi-stadium Digital Twin
 - Cross-city tournament visualization
 - Drone telemetry
 - IoT sensor streaming
 - Apple Vision Pro spatial mode
 - VR Operations Center
 - Live CCTV overlays
 - AI-generated evacuation simulations
 - Multi-user collaborative operations
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Digital Twin Philosophy

"The Digital Twin is not a 3D model—it is the living operational brain of the stadium."

ArenaMind AI transforms every gate, corridor, volunteer, vehicle, and operational signal into an interactive spatial experience where AI, humans, and real-time data collaborate to make faster, safer, and smarter decisions.

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